AI-ENHANCED DOCUMENTATION ANALYSIS IN REGULATORY SAFETY ASSESSMENT OF TRANSPORT PACKAGES

Tobias Gleim
Bundesanstalt für Materialforschung und -prüfung
Berlin, Germany
Email: tobias.gleim@bam.de

Kutlualp Tazefidan
Bundesanstalt für Materialforschung und -prüfung
Berlin, Germany

Frank Wille
Bundesanstalt für Materialforschung und -prüfung
Berlin, Germany

Synopsis

The transport of radioactive material requires regulatory approval based on the package type, as defined by the regulations of the International Atomic Energy Agency (IAEA). These approvals rely on comprehensive Package Design Safety Reports that evaluate mechanical, thermal, shielding, criticality and transport requirements, supported by specifications, inspections, certificates, drawings, and other technical documentations. Such safety reports contain numerous interconnected documents, and even minor changes, such as component modifications, updated material properties or revised regulations, may affect multiple sections. Although all reports follow the same regulatory framework, each package has unique design features, making every safety assessment distinct. Most documentation exists in digital form but remains largely non–machine-interpretable, limiting automated analysis of dependencies across documents. The extended synopsis argues that overcoming these limitations requires moving from simple digitization toward structured knowledge representation. A multi-stage approach begins with foundational AI technologies, including Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), which improve information retrieval but cannot capture the full complexity of safety report interrelationships. Building Knowledge Graphs (KGs) offers the necessary next step by transforming heterogeneous, unstructured, and semi-structured documents into a connected, queryable network. KGs enable precise tracing and visualization of dependencies across datasheets, simulations, experimental results, standards, and regulatory requirements. Such structured representations would allow automatic detection of changes, propagation of effects across related documents and validation of conditions using AI-supported tools, reducing manual workload, and improving safety and consistency. Human error remains a significant factor in drafting and reviewing safety reports. A digital quality infrastructure could reduce the number of iterations and further streamline the overall process. Integrating AI into this workflow has the potential not only to optimize assessments but also to improve their robustness by increasing the interpretability of documentation and thereby enhancing overall safety. This preliminary study examines the readiness and requirements for intelligent documentation analysis systems that support regulatory compliance for transport package safety. By analysing current documentation workflows, it demonstrates how LLM-based tools can interpret complex safety reports and identify critical interdependencies, and why KG-based architectures are essential for managing these dependencies reliably.

1. INTRODUCTION

The transport of radioactive material requires regulatory approval based on the package type, with safety assessments conducted in accordance with the IAEA Regulations for the Safe Transport of Radioactive Material. Safety reports must meet the requirements of SSR-6 [1] and SSG-26 [2], which specify how compliance must be demonstrated. Although the Package Design Safety Report (PDSR) format and SSG-66 [3] provide structural guidance, safety reports still vary widely in content and presentation, as they depend on the specific package design and the practices of designers, manufacturers, and regulatory authorities.

Because packaging designs differ significantly, documentation cannot follow a rigid structure. Regulatory authorities therefore receive highly heterogeneous document sets that include assessments of mechanical, thermal, shielding, criticality and transport requirements, supported by specifications, inspection records, certificates, drawings, and other materials. Experimental test data required under SSR-6 further increase

complexity, and results often appear across multiple documents. These materials combine text, numerical data, drawings, tables, and images, forming dense networks of interdependent information.

Historically, many documents existed only as paper files, and even today digital archives often contain scanned PDFs that are not machine interpretable. OCR improves searchability but cannot capture contextual relationships, leaving experts to manually trace dependencies across documents.

Advances in AI offer potential solutions. Modern systems based on Large Language Models (LLMs) can support semantic search, pattern recognition and the automated linking of related content. An AI-enabled documentation system could integrate heterogeneous materials into a unified knowledge base, enabling automatic detection of changes, propagation of effects across documents and validation of conditions. This would reduce manual cross-checks and improve the consistency and robustness of safety assessments.

However, the use of AI in regulated, safety-critical environments requires consideration of human and organizational factors. Trustworthy AI, transparency and explainability are essential. Human-in-the-Loop approaches ensure that expert judgment remains central, addressing concerns such as accountability, user acceptance and potential long-term deskilling. Actively integrating these sociotechnical aspects strengthens institutional trust, supports regulatory expectations, and enhances the long-term resilience of the safety system.

A digital quality infrastructure (QI-Digital) and AI technologies offer new opportunities for supporting these processes through semantic understanding, multimodal processing, and automated linking between information sources, see FIG. 1. However, their application in safety-critical, regulated environments requires careful consideration of trust, transparency, data protection and explainability.

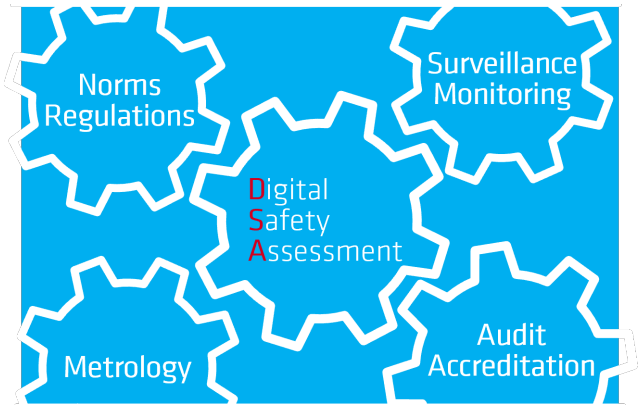


FIG. 1. Digital quality infrastructure for a digital safety assessment.

2. AI TECHNOLOGIES AND THE CHALLENGE OF SMART DOCUMENTATION ANALYSIS IN REGULATED DOMAINS

To effectively address the challenges of manual dependency tracing, an AI-driven system must go beyond basic search capabilities. We propose a multi-layered approach that begins with a foundational baseline and progresses toward a more advanced structured solution.

2.1. Foundational approach: Retrieval-Augmented Generation (RAG)

AI technologies have advanced rapidly and offer strong potential for supporting documentation analysis in safety assessments. However, pretrained Large Language Models (LLMs) lack domain-specific and proprietary knowledge and do not inherently capture the links between related information. Their task-agnostic nature [4] can lead to inaccurate or misleading outputs - so-called hallucinations [6] - which limits their use in safety-critical environments. Using LLMs with sensitive internal data typically requires fine-tuning, which integrates domain knowledge into the model [7]. Yet fine-tuning is resource-intensive, demands continuous monitoring, and raises concerns about data security, especially when cloud-based services provide limited transparency.

RAG has emerged as a foundational approach to address these limitations (Fig. 3). Instead of relying solely on model-internal knowledge, RAG combines a retriever, which identifies relevant documents or passages, with a generator that processes both the user query and the retrieved content [8]. This architecture

allows models to access external, dynamically updated knowledge sources and significantly improves factual accuracy, consistency, and contextual relevance. By grounding responses in retrieved evidence, RAG helps mitigate hallucinations and compensates for missing domain knowledge. Implementing RAG in enterprise settings, however, introduces practical challenges. Retrievers must efficiently search large and frequently updated document repositories [9], while generators must integrate retrieved content coherently with pretrained model knowledge. In addition, organizations must provide secure, performant infrastructure and meet diverse user needs, from intuitive interfaces for end users to flexible configuration options for developers and governance controls for administrators [10].

2.2. Smart documentation analysis in regulated domains

Deploying RAG systems in legally regulated domains such as radioactive material transport, healthcare, finance, or law presents challenges far beyond technical implementation. Stakeholders operate under strict requirements for data protection, security, and traceability, meaning RAG systems must prevent exposure or misuse of sensitive information during retrieval and generation [11]. This necessitates robust access control, anonymization and auditing mechanisms built directly into the system architecture. Because RAG outputs can carry legal, financial or safety implications, expectations for accuracy, consistency and interpretability are significantly higher than in consumer applications. Systems must therefore produce fully auditable results with clear justifications and traceable links back to source documents.

In the nuclear sector, incorrect assumptions can have severe consequences. The transport and storage of radioactive materials require exceptional precision in risk assessment and regulatory compliance. A central aim of this study is to examine whether RAG-based systems can reliably support tasks governed by IAEA regulations or whether current limitations still restrict their use. Enhanced traceability offered by RAG can strengthen expert assessments while improving the efficiency and security of evaluations. This supports BAM's responsibilities in package evaluation and its broader mandate for safety in technology and chemistry.

Despite promising results in other fields, RAG implementation in enterprise and regulated environments faces several core constraints: limited control over retrieval and generation processes, critical for regulatory compliance [12]; difficulties processing large, heterogeneous enterprise knowledge bases [13]; insufficient transparency, hindering trust and accountability in mission-critical tasks [13]; integration challenges related to security, governance, and data management.

These limitations highlight the need for specialized RAG solutions designed specifically for sensitive and regulated domains. Such systems must prioritize compliance, reliability, and transparency to enable their safe and effective adoption.

2.3. Machine-readable and machine-understandable documents

The heterogeneous nature of safety documentation, ranging from text to drawings, diagrams, tables, and sensor data requires multimodal processing. OCR alone cannot interpret technical drawings or complex layouts. Vision AI and layout-aware models can extract meaningful information from diagrams, tables, and graphical elements. Reprocessing all documents with modern AI-based parsing enables semantic search and cross-referencing across diverse materials.

2.4. The need for structured knowledge: Knowledge Graphs

To capture the intricate dependencies within safety assessments, information must be transformed into structured, machine-interpretable form. KGs model entities such as components, materials, certificates or regulatory clauses and their semantic relations. Building such graphs requires layout-aware parsing, named entity recognition and relation extraction. A KG consolidates fragmented information into a unified representation and provides a foundation for advanced reasoning.

2.5. Advanced question answering with Graph RAG

Graph RAG extends classical RAG by operating directly on KG structures. It supports multi-hop reasoning across interconnected entities and enables traceable, explainable responses. For example, a change in

boundary conditions, such as temperature, can be propagated through the KG to identify affected components, associated limits, and relevant regulatory requirements. This approach provides transparency essential for safety-critical assessments.

2.6. Human factors, trust and transparency

The integration of AI in regulated domains depends on users' trust and understanding of system behaviour. Explainable AI (XAI), human-in-the-loop approaches and clear accountability structures are essential. Over-automation may lead to de-skilling or loss of domain expertise, cf. [14]. Therefore, AI must support rather than replace expert judgement and be designed to maintain long-term organizational competence.

3. IMPLEMENTATION AND OUTLOOK

A central challenge for AI-supported documentation systems is the pre-processing of heterogeneous data. Much of the relevant information exists as PDFs or other formats that are not machine-readable or sufficiently structured for automated analysis. These files must first be converted into machine-interpretable formats using NLP tools that extract and restructure content for KG construction, see FIG. 2. Semantic technologies such as ontologies and metadata schemas further improve interpretability and interoperability, as shown in initiatives like Plattform Material Digital and FAIR data practices, cf. [15].

Beyond software and hardware, an integrated modular platform is needed to map the entire workflow from data ingestion to output generation. Replacing LLMs or other AI components must not disrupt information flows. To meet varying regulatory and security requirements, the platform should support both local and cloud-based deployment. A model manager coordinates the flexible use of LLMs, while a multi-agent system selects suitable models and data sources depending on task and user prompt, as illustrated in FIG. 2. Because system configurations and technologies continue to evolve, the architecture must remain adaptable.

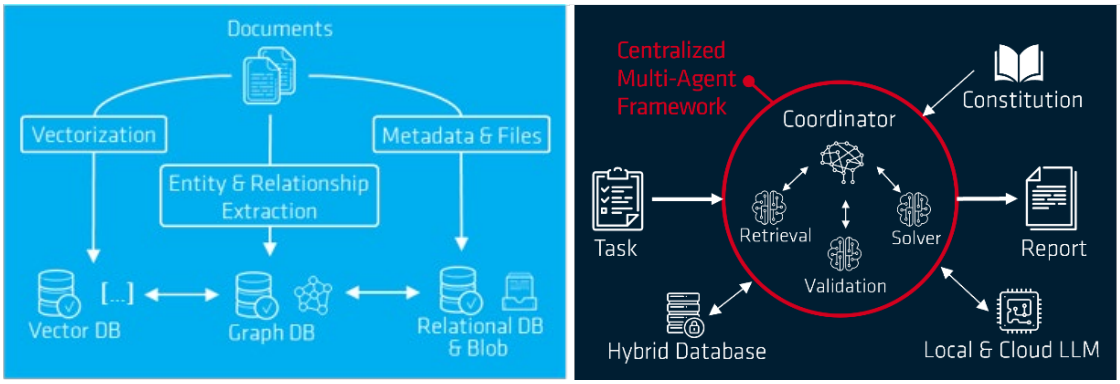


FIG. 2. Database for safety assessment and knowledge-grounded document analysis.

Depending on the deployment strategy, users may maintain their own infrastructure or rely on external providers. Introducing AI systems typically requires initial investment and personnel effort; the goal is improved safety, transparency, and consistency rather than immediate cost savings. Over time, infrastructure maturity and system design determine whether efficiency gains such as faster evaluations and more consistent documentation can be achieved.

In a human-centred system, these developments must align with user needs as well as technical criteria. Evaluation metrics should consider not only output quality but also usability and process impact. Improvements may result from better prompts, enhanced retrieval, or more structured datasets, ranging from file organization to ontology-based structuring. Reliable version control is essential to ensure that current document revisions are analysed.

As data pipelines mature, machine-interpretable Smart Standards will become increasingly important. Once regulatory clauses are available as executable logic, they can be integrated into the KG alongside monitoring data and lifecycle certificates. A deterministic validation engine can then automate compliance

checks, while agentic AI components identify regulatory changes and generate updates for expert review. This would shift documentation workflows from reactive to continuously updated safety assurance.

BAM is advancing these technologies in several internal research projects and, within the current initiative on radioactive material transport, will continue to examine the outlined questions step by step.

4. CONCLUSIONS

The extended synopsis examines the potential of AI-supported documentation and analysis systems in regulated safety environments, using the transport of radioactive materials as an illustrative example. Current documentation processes in such domains are highly complex, fragmented, and labour-intensive, with limited support for semantic linking or automated evaluation. Although technologies such as LLMs and RAG provide an important foundation, they do not fully address one of the central issues: the strong interdependencies between documents. A key step toward overcoming this limitation is the development of a KG that integrates the heterogeneous document corpus into a connected, machine-readable structure. When combined with advanced Graph-RAG techniques, such a KG enables multi-step, explainable querying, an essential capability for safety-critical applications.

Successful implementation, however, depends on more than technical feasibility. Stakeholder feedback shows considerable interest in AI-driven approaches, but also highlights persistent concerns regarding cybersecurity, resource requirements, trust, and day-to-day usability. At present, no commercial solutions are able to analyse complex, interlinked documentation while simultaneously meeting the strict data protection and transparency requirements of regulated sectors. Addressing these challenges requires not only technological advances but also organisational and human-centred measures. This includes building trust, ensuring explainability, and designing systems that operate transparently and reliably in safety-relevant contexts.

REFERENCES

[1] IAEA, Specific Safety Requirements SSR-6 Rev. 1, Regulations for the Safe Transport of Radioactive Material, 2018 Edition, DOI: https://doi.org/10.61092/iaea.ur52-my9o

[2] IAEA, Specific Safety Guide SSG-26 Rev. 1, Advisory Material for the IAEA Regulations for the Safe Transport of Radioactive Material (2018 Edition), DOI: https://doi.org/10.61092/iaea.qz7d-jiym.

[3] IAEA, Specific Safety Guide SSG-66, Format and Content of the Package Design Safety Report for the Transport of Radioactive Material, URL: https://www-pub.iaea.org/MTCD/Publications/PDF/PUB1980_web.pdf.

[4] Brown, T. B. et. al., Language Models are Few-Shot Learners, arXiv: https://arxiv.org/abs/2005.14165, 2020.

[5] Touvron, H. et. al., LLaMA: Open and Efficient Foundation Language Models, arXiv: https://arxiv.org/pdf/2302.13971, 2023.

[6] Shuster, K. et. al., Retrieval Augmentation Reduces Hallucination in Conversation, arXiv: https://arxiv.org/abs/2104.07567, 2021.

[7] Junlong, L. et. al., DiT: Self-supervised Pre-training for Document Image Transformer, arXiv: https://doi.org/10.1145/3503161.3547911, 2022.
www.webpage.com/exact-subpage-being-cited

[8] Lewis, P. et. al., Retrieval-augmented generation for knowledge-intensive nlp tasks, arXiv: https://arxiv.org/pdf/2005.11401, 2020.

[9] Shao, Z. et. al., Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy, arXiv: https://arxiv.org/abs/2305.15294, 2023.

[10] Han, Y. et. al., A Comprehensive Survey on Vector Database: Storage and Retrieval Technique, Challenge, arXiv: https://arxiv.org/pdf/2310.11703, 2023.

[11] Arrieta, A. B. et. al., Explainable Artificial Intelligence (XAI): Concepts, taxonomies,opportunities and challenges toward responsible AI, arXiv: https://arxiv.org/abs/1910.10045, 2020.

[12] Anderljung, M. et. al., Frontier AI Regulation: Managing Emerging Risks to Public Safety, arXiv: https://arxiv.org/pdf/2307.03718, 2023.

[13] Ahmad, A. A.-S., Andras, P., Scalability analysis comparisons of cloud-based software services, Springer Open: https://doi.org/10.1186/s13677-019-0134-y, 2019.

[14] Shneiderman, B., Human-Centered Artificial Intelligence: Reliable, Safe & Trustworthy, International Journal of Human–Computer Interaction, vol. 36, no. 6, pp. 495-504: arXiv: https://arxiv.org/abs/2002.04087v1, 2020.

[15] Schilling, M. et al., FAIR and Structured Data: A Domain Ontology Aligned with Standard-Compliant Tensile Testing, Advanced Engineering Materials: DOI: https://doi.org/10.1002/adem.202400138, 2024.